

Instructions:

You will work on these exercises during the practical sessions in the second week of the course.

Below is an indication of the exercises that we will cover (at least) in each session. We encourage you try them before the session.

- First session (Wednesday/Thursday): exercises 1, 2.
- Second session (Friday): exercises 4, 5.

Exercise 1: MAP Solution for Linear Regression

In this exercise, you will solve and practice with the maximum a posteriori (MAP) estimator for linear regression with basis functions. We assume N training vectors $\{\mathbf{x}_n\}_{n=1}^N$, each mapped to a feature vector

$$\phi_n := \phi(\mathbf{x}_n) = (\phi_0(\mathbf{x}_n), \phi_1(\mathbf{x}_n), \dots, \phi_{M-1}(\mathbf{x}_n))^T,$$

using basis functions $\phi_j(\mathbf{x})$ for $j = 0, \dots, M-1$, where $\phi_0(\mathbf{x}) = 1$ is a bias basis. The training set consists of input-output pairs $(\mathbf{x}_n, t_n)$. The regression prediction is

$$y(\mathbf{x}_n, \mathbf{w}) = \mathbf{w}^T \phi_n,$$

and the data samples are independent.

For each scalar target t_n , the likelihood is Gaussian:

$$p(t_n | \phi_n, \mathbf{w}, \beta) = \mathcal{N}(t_n | \mathbf{w}^T \phi_n, \beta^{-1}),$$

where $\beta > 0$ is the observation-noise precision, i.e., the inverse of the noise variance. Thus, β^{-1} is the variance of each target around its predicted mean. The weights are independent a priori, with a Gaussian prior for each scalar weight:

$$p(w_j | \alpha) = \mathcal{N}(w_j | 0, \alpha^{-1}), \quad j = 0, \dots, M-1.$$

Here, $\alpha > 0$ is the prior precision of each weight, so α^{-1} is its prior variance. The MAP solution for the weights is

$$\mathbf{w}_{\text{MAP}} = (\Phi^T \Phi + \lambda \mathbf{I})^{-1} \Phi^T \mathbf{t},$$

where $\mathbf{t} = (t_1, \dots, t_N)^T \in \mathbb{R}^N$ is the vector of observed targets and $\mathbf{I} \in \mathbb{R}^{M \times M}$ is the identity matrix. The design matrix is

$$\Phi = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{pmatrix}.$$

Derive the MAP solution by answering the following questions.

- (a) Derive the likelihood $p(\mathcal{D} | \mathbf{w})$ for the full training set as a product over the N observations. Then express it as a multivariate Gaussian density in vector/matrix form, identifying its mean vector and covariance matrix. Compute the log-likelihood $\ln p(\mathcal{D} | \mathbf{w})$ in both forms.

Solution:

By independence,

$$p(\mathcal{D} | \mathbf{w}) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^\top \phi_n, \beta^{-1}) = \prod_{n=1}^N \frac{\beta^{1/2}}{(2\pi)^{1/2}} \exp \left[-\frac{\beta}{2} (t_n - \mathbf{w}^\top \phi_n)^2 \right].$$

Therefore,

$$p(\mathcal{D} | \mathbf{w}) = \frac{\beta^{N/2}}{(2\pi)^{N/2}} \exp \left[-\frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^\top \phi_n)^2 \right].$$

Using the target vector $\mathbf{t}$ and design matrix Φ defined above, whose n -th row is $\phi_n^\top$, we obtain

$$p(\mathcal{D} | \mathbf{w}) = \frac{\beta^{N/2}}{(2\pi)^{N/2}} \exp \left[-\frac{\beta}{2} (\mathbf{t} - \Phi \mathbf{w})^\top (\mathbf{t} - \Phi \mathbf{w}) \right] = \mathcal{N}(\mathbf{t} | \Phi \mathbf{w}, \beta^{-1} \mathbf{I}).$$

Taking logarithms of the product form gives

$$\ln p(\mathcal{D} | \mathbf{w}) = \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) - \frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^\top \phi_n)^2.$$

Equivalently, in matrix form,

$$\ln p(\mathcal{D} | \mathbf{w}) = \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) - \frac{\beta}{2} (\mathbf{t} - \Phi \mathbf{w})^\top (\mathbf{t} - \Phi \mathbf{w}).$$

The first two terms are constant with respect to $\mathbf{w}$.

- (b) Derive the joint prior $p(\mathbf{w})$ as a product over the M weights. Then express it as a multivariate Gaussian density, identifying its mean vector and covariance matrix. Write out the density explicitly and compute $\ln p(\mathbf{w})$.

Solution:

By independence of the weights under the prior, suppressing the conditioning on the fixed precision α ,

$$p(\mathbf{w}) = \prod_{j=0}^{M-1} p(w_j | \alpha) = \prod_{j=0}^{M-1} \frac{\alpha^{1/2}}{(2\pi)^{1/2}} \exp \left(-\frac{\alpha}{2} w_j^2 \right) = \frac{\alpha^{M/2}}{(2\pi)^{M/2}} \exp \left(-\frac{\alpha}{2} \sum_{j=0}^{M-1} w_j^2 \right).$$

Since $\sum_{j=0}^{M-1} w_j^2 = \mathbf{w}^\top \mathbf{w}$, this is a multivariate Gaussian with mean $\mathbf{0} \in \mathbb{R}^M$ and covariance $\alpha^{-1} \mathbf{I}_M$:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}) = \frac{\alpha^{M/2}}{(2\pi)^{M/2}} \exp \left(-\frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} \right).$$

Taking logarithms gives

$$\ln p(\mathbf{w}) = \frac{M}{2} \ln \alpha - \frac{M}{2} \ln(2\pi) - \frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} = C - \frac{\alpha}{2} \mathbf{w}^\top \mathbf{w},$$

where C does not depend on $\mathbf{w}$.

- (c) Write down an expression for the posterior $p(\mathbf{w} | \mathcal{D})$ by applying Bayes' rule. You do not need to write out the explicit Gaussian densities. Show that the evidence requires an integral, which

you do not need to solve analytically, and then replace it with the corresponding probability distribution.

Solution:

Bayes' rule gives

$$p(\mathbf{w} \mid \mathcal{D}) = \frac{p(\mathbf{w})p(\mathcal{D} \mid \mathbf{w})}{p(\mathcal{D})}.$$

In product form,

$$p(\mathbf{w} \mid \mathcal{D}) = \frac{\mathcal{N}(\mathbf{w} \mid \mathbf{0}, \alpha^{-1}\mathbf{I}) \prod_{n=1}^N \mathcal{N}(t_n \mid \mathbf{w}^\top \phi_n, \beta^{-1})}{\int \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \alpha^{-1}\mathbf{I}) \prod_{n=1}^N \mathcal{N}(t_n \mid \mathbf{w}^\top \phi_n, \beta^{-1}) d\mathbf{w}}.$$

Equivalently, in matrix form,

$$p(\mathbf{w} \mid \mathcal{D}) = \frac{\mathcal{N}(\mathbf{w} \mid \mathbf{0}, \alpha^{-1}\mathbf{I}) \mathcal{N}(\mathbf{t} \mid \Phi \mathbf{w}, \beta^{-1}\mathbf{I})}{\int \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \alpha^{-1}\mathbf{I}) \mathcal{N}(\mathbf{t} \mid \Phi \mathbf{w}, \beta^{-1}\mathbf{I}) d\mathbf{w}}.$$

The denominator is the evidence,

$$p(\mathbf{t} \mid \Phi, \alpha, \beta),$$

so

$$p(\mathbf{w} \mid \mathcal{D}) = \frac{\mathcal{N}(\mathbf{w} \mid \mathbf{0}, \alpha^{-1}\mathbf{I}) \mathcal{N}(\mathbf{t} \mid \Phi \mathbf{w}, \beta^{-1}\mathbf{I})}{p(\mathbf{t} \mid \Phi, \alpha, \beta)}.$$

- (d) Compute the log-posterior for both likelihood expressions from the first question (product over N and matrix form). Collect all terms independent of $\mathbf{w}$ into a constant C . Which parts do not depend on $\mathbf{w}$? Why is finding the MAP simpler than finding the full posterior distribution?

Solution:

Before taking the logarithm, the full expression for the posterior using the product-form likelihood is:

$$p(\mathbf{w} \mid \mathcal{D}) = \frac{1}{p(\mathbf{t} \mid \Phi, \alpha, \beta)} \left(\frac{\alpha^{M/2}}{(2\pi)^{M/2}} \exp\left(-\frac{\alpha}{2} \mathbf{w}^\top \mathbf{w}\right) \right) \left(\frac{\beta^{N/2}}{(2\pi)^{N/2}} \exp\left[-\frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^\top \phi_n)^2\right] \right) \\ \propto \exp\left(-\frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} - \frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^\top \phi_n)^2\right).$$

Taking the logarithm, we obtain:

$$\ln p(\mathbf{w} \mid \mathcal{D}) = -\frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} - \frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^\top \phi_n)^2 + C.$$

In matrix form, this is:

$$\ln p(\mathbf{w} \mid \mathcal{D}) = -\frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} - \frac{\beta}{2} (\mathbf{t} - \Phi \mathbf{w})^\top (\mathbf{t} - \Phi \mathbf{w}) + C.$$

The Gaussian normalization constants and the evidence $p(\mathbf{t} \mid \Phi, \alpha, \beta)$ do not depend on $\mathbf{w}$, so they are absorbed into C . Finding the MAP is simpler than finding the full posterior because MAP only requires maximizing with respect to $\mathbf{w}$, so all constants independent of $\mathbf{w}$ can be ignored. The full posterior requires computing the normalizing evidence.

- (e) Solve for $\mathbf{w}_{\text{MAP}}$ by taking the derivative of the log-posterior with respect to $\mathbf{w}$, setting it to zero, and solving for $\mathbf{w}$. Do this for both forms of the log-posterior.

Solution:

Starting from the sum form,

$$\ln p(\mathbf{w} \mid \mathcal{D}) = -\frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} - \frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^\top \phi_n)^2 + C.$$

Differentiating and setting the gradient to zero gives

$$-\alpha \mathbf{w} + \beta \sum_{n=1}^N \phi_n (t_n - \phi_n^\top \mathbf{w}) = \mathbf{0}.$$

Hence

$$\left(\alpha \mathbf{I} + \beta \sum_{n=1}^N \phi_n \phi_n^\top \right) \mathbf{w} = \beta \sum_{n=1}^N t_n \phi_n,$$

and therefore

$$\mathbf{w}_{\text{MAP}} = \left(\alpha \mathbf{I} + \beta \sum_{n=1}^N \phi_n \phi_n^\top \right)^{-1} \beta \sum_{n=1}^N t_n \phi_n.$$

In matrix form, expanding the log-posterior and dropping constants gives

$$\ln p(\mathbf{w} \mid \mathcal{D}) = -\frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} - \frac{\beta}{2} \mathbf{w}^\top \Phi^\top \Phi \mathbf{w} + \beta \mathbf{w}^\top \Phi^\top \mathbf{t} + C.$$

Setting the derivative to zero,

$$-\alpha \mathbf{w} - \beta \Phi^\top \Phi \mathbf{w} + \beta \Phi^\top \mathbf{t} = \mathbf{0}.$$

Thus

$$\mathbf{w}_{\text{MAP}} = (\alpha \mathbf{I} + \beta \Phi^\top \Phi)^{-1} \beta \Phi^\top \mathbf{t} = \left(\frac{\alpha}{\beta} \mathbf{I} + \Phi^\top \Phi \right)^{-1} \Phi^\top \mathbf{t}.$$

Therefore $\lambda = \alpha/\beta$.

- (f) The prior for $\mathbf{w}$ assumes the same distribution for each entry, including w_0 , which is multiplied by the basis function $\phi_0(\mathbf{x}) = 1$. What is the role of w_0 and $\phi_0(\mathbf{x})$? Why should we avoid placing the same penalty/prior on this basis? Rewrite $p(\mathbf{w})$ so that w_0 has its own prior/penalty.

Solution:

The constant basis function acts as a bias or offset for the regression problem. If we use the same prior for this weight as for the others, we are assuming that the offset from the y-axis should somehow be penalised. This does not make too much sense a priori, so instead we use a different precision for this basis function, i.e. $\alpha_0 \ll \alpha$, while using α for all the others.

Exercise 2: Absolute Error for linear regression

Recall the least-squares error function for Linear Regression:

$$E(\mathbf{w}) = \frac{1}{2}(y - \mathbf{w}^\top \mathbf{x})^2.$$

This is the error associated with a single training sample with input $\mathbf{x}$ and target value y .

An alternative is the absolute error:

$$J(\mathbf{w}) = |y - \mathbf{w}^\top \mathbf{x}|.$$

- (a) Which of the two error functions is more robust to outliers? Why?

Solution:

The absolute error is more robust. For residual $r = y - \mathbf{w}^\top \mathbf{x}$, $E = r^2$, whereas $J = |r|$ grows linearly. Thus large residuals have bigger impact on E than on J .

- (b) Derive the gradient-descent update rule for this new objective function. You can assume $y - \mathbf{w}^\top \mathbf{x} \neq 0$.

Solution:

Let

$$u = y - \mathbf{w}^\top \mathbf{x} = y - \sum_{j=0}^n w_j x_j.$$

Then the absolute error is

$$J(\mathbf{w}) = |u|.$$

For a particular weight w_i , we use the chain rule, obtaining that for $u \neq 0$:

$$\frac{\partial J}{\partial w_i} = \frac{u}{|u|} \frac{\partial u}{\partial w_i}$$

Note: to see where $\frac{\partial |u|}{\partial u} = \frac{u}{|u|}$ comes from, write $|u| = \sqrt{u^2}$ and apply the chain rule:

$$\frac{\partial |u|}{\partial u} = \frac{\partial}{\partial u} \sqrt{u^2} = \frac{1}{2\sqrt{u^2}} \cdot 2u = \frac{u}{\sqrt{u^2}} = \frac{u}{|u|}.$$

Since

$$\frac{\partial u}{\partial w_i} = \frac{\partial}{\partial w_i} \left(y - \sum_{j=0}^n w_j x_j \right) = -x_i,$$

we get

$$\frac{\partial J}{\partial w_i} = -\frac{y - \mathbf{w}^\top \mathbf{x}}{|y - \mathbf{w}^\top \mathbf{x}|} x_i.$$

Thus, in vector form,

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{y - \mathbf{w}^\top \mathbf{x}}{|y - \mathbf{w}^\top \mathbf{x}|} \mathbf{x},$$

and the update rule is

$$\mathbf{w} \leftarrow \mathbf{w} + \eta \frac{y - \mathbf{w}^\top \mathbf{x}}{|y - \mathbf{w}^\top \mathbf{x}|} \mathbf{x},$$

where $\eta > 0$ is the learning rate.

Equivalently, using the sign function,

$$\mathbf{w} \leftarrow \mathbf{w} + \eta \operatorname{sign}(y - \mathbf{w}^\top \mathbf{x}) \mathbf{x}.$$

- (c) What happens when $y - \mathbf{w}^\top \mathbf{x} = 0$? What is a reasonable way to define the update rule in this case?

Solution:

When $y - \mathbf{w}^\top \mathbf{x} = 0$, the absolute value is not differentiable. A reasonable standard choice is taking the gradient as zero:

$$\mathbf{w} \leftarrow \mathbf{w},$$

so all weights are left unchanged for a perfectly predicted sample.

Exercise 3: Bayesian Linear Regression with Laplace Prior

Exam question

Consider the linear regression model

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$

where $\mathbf{X} \in \mathbb{R}^{N \times D}$ is a data matrix consisting of N i.i.d. samples, $\mathbf{y} \in \mathbb{R}^N$ is the vector of targets, and $\mathbf{w} \in \mathbb{R}^D$ is the vector of coefficients. Each coefficient w_d , for $d \in \{1, \dots, D\}$, has an independent Laplace prior

$$p(w_d) = \frac{\alpha}{2} \exp(-\alpha |w_d|), \quad \alpha > 0.$$

Show that the optimization problem for the MAP estimator is

$$\hat{\mathbf{w}}_{\text{MAP}} = \arg \min_{\mathbf{w} \in \mathbb{R}^D} \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1.$$

Solution:

The likelihood is Gaussian:

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) = \mathcal{N}(\mathbf{y} \mid \mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N).$$

Writing out the density gives

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) = (2\pi\sigma^2)^{-N/2} \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2\right).$$

Since the coefficients are independent under the prior,

$$p(\mathbf{w}) = \prod_{d=1}^D p(w_d) = \prod_{d=1}^D \frac{\alpha}{2} \exp(-\alpha |w_d|).$$

Therefore,

$$p(\mathbf{w}) = \left(\frac{\alpha}{2}\right)^D \exp\left(-\alpha \sum_{d=1}^D |w_d|\right) = \left(\frac{\alpha}{2}\right)^D \exp(-\alpha \|\mathbf{w}\|_1).$$

By Bayes' rule,

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) p(\mathbf{w}).$$

Taking logarithms and collecting all terms independent of $\mathbf{w}$ into a constant C gives

$$\ln p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = -\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 - \alpha \|\mathbf{w}\|_1 + C.$$

Maximizing the posterior is equivalent to minimizing the negative log-posterior. Dropping constants independent of $\mathbf{w}$, we obtain

$$\hat{\mathbf{w}}_{\text{MAP}} = \arg \min_{\mathbf{w} \in \mathbb{R}^D} \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1.$$

Thus, a Laplace prior on the weights leads to an ℓ_1 -regularized least-squares objective.

Exercise 4: Posterior Uncertainty in Bayesian Linear Regression

We consider the Bayesian linear regression model presented in the lectures and in Section 3.3. of Bishop [1], where the posterior precision matrix after observing N data points is

$$\mathbf{S}_N^{-1} = \alpha \mathbf{I} + \beta \sum_{n=1}^N \phi(\mathbf{x}_n) \phi(\mathbf{x}_n)^\top,$$

where $\alpha, \beta > 0$ are the prior and observation precisions,¹ respectively. The variance of the predictive distribution for the target at a new input $\mathbf{x}$ is

$$\sigma_N^2(\mathbf{x}) = \frac{1}{\beta} + \phi(\mathbf{x})^\top \mathbf{S}_N \phi(\mathbf{x}).$$

The term $1/\beta$ represents the noise on the data, while $\phi(\mathbf{x})^\top \mathbf{S}_N \phi(\mathbf{x})$ is the uncertainty associated with the model weights.

- (a) Show that $\mathbf{S}_N$ is positive definite.

Solution:

It is enough to show that $\mathbf{S}_N^{-1}$ is positive definite, because the inverse of a positive definite matrix is also positive definite. For any nonzero vector $\mathbf{z}$,

$$\mathbf{z}^\top \mathbf{S}_N^{-1} \mathbf{z} = \mathbf{z}^\top \left(\alpha \mathbf{I} + \beta \sum_{n=1}^N \phi(\mathbf{x}_n) \phi(\mathbf{x}_n)^\top \right) \mathbf{z}.$$

Expanding this expression gives

$$\mathbf{z}^\top \mathbf{S}_N^{-1} \mathbf{z} = \alpha \|\mathbf{z}\|^2 + \beta \sum_{n=1}^N \left(\mathbf{z}^\top \phi(\mathbf{x}_n) \right)^2.$$

Since $\alpha > 0$ and $\mathbf{z} \neq \mathbf{0}$, the first term is strictly positive. Since $\beta > 0$, the second term is nonnegative. Therefore

$$\mathbf{z}^\top \mathbf{S}_N^{-1} \mathbf{z} > 0$$

for all $\mathbf{z} \neq \mathbf{0}$, so $\mathbf{S}_N^{-1}$ is positive definite. Hence $\mathbf{S}_N$ is positive definite as well.

- (b) Show that adding one more training point $\mathbf{x}_{N+1}$ cannot increase the uncertainty of the predictive distribution, i.e.,

$$\sigma_{N+1}^2(\mathbf{x}) \leq \sigma_N^2(\mathbf{x}).$$

Hint: Use the following matrix identity: $(\mathbf{M} + \mathbf{v}\mathbf{v}^\top)^{-1} = \mathbf{M}^{-1} - \frac{(\mathbf{M}^{-1}\mathbf{v})(\mathbf{v}^\top \mathbf{M}^{-1})}{1 + \mathbf{v}^\top \mathbf{M}^{-1} \mathbf{v}}$

¹The precision is the inverse of the variance.

Solution:

After adding the data point $\mathbf{x}_{N+1}$, the posterior precision becomes

$$\mathbf{S}_{N+1}^{-1} = \mathbf{S}_N^{-1} + \beta \phi(\mathbf{x}_{N+1}) \phi(\mathbf{x}_{N+1})^\top.$$

Equivalently,

$$\mathbf{S}_{N+1}^{-1} = \mathbf{S}_N^{-1} + \left(\sqrt{\beta} \phi(\mathbf{x}_{N+1}) \right) \left(\sqrt{\beta} \phi(\mathbf{x}_{N+1}) \right)^\top.$$

Applying the matrix identity with

$$\mathbf{M} = \mathbf{S}_N^{-1}, \quad \mathbf{v} = \sqrt{\beta} \phi(\mathbf{x}_{N+1}),$$

gives

$$\mathbf{S}_{N+1} = \mathbf{S}_N - \frac{\beta \mathbf{S}_N \phi(\mathbf{x}_{N+1}) \phi(\mathbf{x}_{N+1})^\top \mathbf{S}_N}{1 + \beta \phi(\mathbf{x}_{N+1})^\top \mathbf{S}_N \phi(\mathbf{x}_{N+1})}.$$

Therefore,

$$\begin{aligned} \sigma_N^2(\mathbf{x}) - \sigma_{N+1}^2(\mathbf{x}) &= \phi(\mathbf{x})^\top (\mathbf{S}_N - \mathbf{S}_{N+1}) \phi(\mathbf{x}) \\ &= \frac{\beta \phi(\mathbf{x})^\top \mathbf{S}_N \phi(\mathbf{x}_{N+1}) \phi(\mathbf{x}_{N+1})^\top \mathbf{S}_N \phi(\mathbf{x})}{1 + \beta \phi(\mathbf{x}_{N+1})^\top \mathbf{S}_N \phi(\mathbf{x}_{N+1})} \\ &= \frac{[\phi(\mathbf{x})^\top \mathbf{S}_N \phi(\mathbf{x}_{N+1})]^2}{\beta^{-1} + \phi(\mathbf{x}_{N+1})^\top \mathbf{S}_N \phi(\mathbf{x}_{N+1})}. \end{aligned}$$

The numerator is a square, and the denominator is positive because $\beta > 0$ and $\mathbf{S}_N$ is positive definite by the previous part. Hence

$$\sigma_N^2(\mathbf{x}) - \sigma_{N+1}^2(\mathbf{x}) \geq 0,$$

and therefore

$$\sigma_{N+1}^2(\mathbf{x}) \leq \sigma_N^2(\mathbf{x}).$$

Thus, adding a data point cannot increase the posterior contribution to predictive uncertainty.

Adapted from Chapter 3 of Pattern Recognition and Machine Learning [1].

Exercise 5: Bias and Variance

Explain what effect the following operations will have on the bias and variance of your model. Fill in one of “increases”, “decreases”, or “no change” in each of the cells.

Operation	Bias	Variance	Justification
Regularizing the weights in a linear/logistic regression model			
Pruning a decision tree (to a certain depth, for example)			
Using dropout to train a deep neural network			
Increasing k in k -nearest neighbor models			
Averaging multiple decision trees in a random forest			
Increasing the degree of polynomial regression			
Increasing the number of hidden units in a neural network			
Using fewer training examples			

Solution:

Operation	Bias	Variance	Justification
Regularizing the weights in a linear/logistic regression model	increases	decreases	Regularization constrains the fitted weights, making the model less flexible but more stable.
Pruning a decision tree (to a certain depth, for example)	increases	decreases	Pruning removes splits, so the tree captures fewer details but varies less across datasets.
Using dropout to train a deep neural network	increases	decreases	Dropout acts as regularization by preventing hidden units from co-adapting too strongly.
Increasing k in k -nearest neighbor models	increases	decreases	Larger neighborhoods smooth predictions, reducing sensitivity to individual training points.
Averaging multiple decision trees in a random forest	no change	decreases	Averaging mainly reduces variability across individual trees while keeping their average flexibility similar.
Increasing the degree of polynomial regression	decreases	increases	Higher-degree polynomials are more flexible and can adapt more strongly to the training sample.
Increasing the number of hidden units in a neural network	decreases	increases	More hidden units increase model capacity, allowing more complex functions but also more overfitting.
Using fewer training examples	no change	increases	The type of model stays the same, but with fewer examples the fitted model depends more on which examples were sampled.

References

- [1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006.